



Information Technology Institute (ITI)
Analog IC Design

Design Challenge

Self-Biased Sub-1V Bandgap Reference Circuit

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1 Design Specifications and Approach

1.1 Given Specs

Parameter	Value
Technology	65 nm
Supply voltage (V_{DD})	1.2 V
Corners	Temp: -40 to 125 °C, Process: SS, SF, FS, FF
Output voltage (V_{OUT})	0.8 V
Total bias current	$< 10 \mu\text{A}$
Phase margin	$> 60^\circ$

Table 1: Design challenge specifications

2 Part 1: BGR Core

2.1 Hand Analysis – Output Voltage Equation

The amplifier forces (R2–Q2) to equal (R1–Q1). Since V_{EB1} (a CTAT quantity, $b \approx -1.5$ to -2 mV/°C) appears directly across R1:

$$I_1 = \frac{V_{EB1}}{R_1} \quad (1)$$

Q2 has area $N \times$ larger than Q1, so at the same current density the loop forces $\Delta V_{EB} = V_{EB1} - V_{EB2} = V_T \ln(N)$ across R2 (a PTAT quantity):

$$I_2 = \frac{V_T \ln(N)}{R_2} \quad (2)$$

MP3 mirrors the sum of both branch currents into R3, giving:

$$V_{OUT} = R_3 \left(\frac{V_{EB1}}{R_1} + \frac{V_T \ln N}{R_2} \right) = R_3(I_1 + I_2) \quad (3)$$

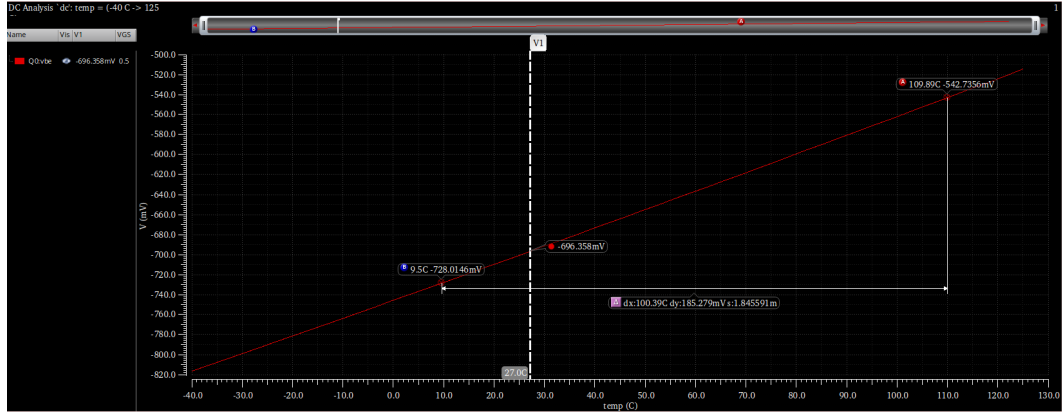
2.2 Temperature Cancellation Condition

Differentiating V_{OUT} w.r.t. T and setting $dV_{OUT}/dT = 0$:

$$\left. \frac{V_{EB1}}{R_1} \right|_{\text{contributes}} + \frac{d}{dT} \left(\frac{V_{EB1}}{R_1} \right) + \frac{d}{dT} \left(\frac{V_T \ln N}{R_2} \right) = 0 \implies \frac{R_2}{R_1} = \frac{\ln(N) \cdot (dV_T/dT)}{|dV_{EB1}/dT|} \quad (4)$$

with $dV_T/dT = k/q = V_T/T \approx 0.0862$ mV/K at $T = 300$ K.

Before sizing, I swept a diode-connected PNP from -40 °C to 125 °C at a current density corresponding to $I_C \approx 2 \mu\text{A}$ and extracted $V_{EB}(T)$ and its slope.

Figure 1: PNP characterization: $V_{EB}(T)$ sweep used to extract dV_{EB}/dT

Parameter	Extracted Value
$V_{EB1} @ 27^\circ\text{C}, I_C = 2\mu\text{A}$	$\approx 0.7\text{ V}$
dV_{EB1}/dT	$\approx -1.8\text{ mV}/^\circ\text{C}$

Table 2: PNP parameters used in the hand analysis (to be confirmed/updated from simulation)

2.3 R_1, R_2, R_3

Using $N = 24$

Step 1 – ΔV_{EB} at $N = 24$:

$$\Delta V_{EB} = V_T \ln(24) = 25.85\text{mV} \times 3.178 = 82.16\text{ mV} \quad (5)$$

Step 2 – Check the R_2/R_1 ratio against the schematic's own values (sanity check):

$$\frac{R_2}{R_1} = \frac{\ln(24) \times 0.0862}{1.} = \frac{0.2740}{1.7} = 0.1522 \quad (6)$$

Step 3 – Pick I_1 from the current budget. the current will be (CTAT + PTAT + their sum mirrored into the output). To leave margin under $10\mu\text{A}$ for the amplifier and startup circuit, I selected that every branch of the core will have a current of $2\mu\text{A}$ and for the error amplifier and it will have $1.5\mu\text{A}$ so the total current is $7.5\mu\text{A}$. $I_1 + I_2 \approx 2\mu\text{A}$. I choose $I_1 = 1\mu\text{A}$.

$$R_1 = \frac{V_{EB1}}{I_1} = \frac{0.7\text{ V}}{1.15\mu\text{A}} = 608\text{ k}\Omega \approx 600\text{ k}\Omega \quad (7)$$

$$R_2 = 0.1522 \times R_1 = 0.1522 \times 608\text{k} \approx 92.6\text{ k}\Omega \approx 90\text{ k}\Omega \quad (8)$$

$$I_2 = \frac{\Delta V_{EB}}{R_2} = \frac{82.16\text{ mV}}{90\text{ k}\Omega} = 0.913\mu\text{A} \quad (9)$$

Step 4 – Solve R_3 for $V_{OUT} = 0.8\text{ V}$:

$$R_3 = \frac{V_{OUT}}{I_1 + I_2} = \frac{0.8\text{ V}}{2\mu\text{A}} \approx 400\text{ k}\Omega \quad (10)$$

Total core current:

$$I_{core} = I_1 + I_2 + (I_1 + I_2) = 6\mu\text{A} \quad (11)$$

This leaves $\approx 4\mu\text{A}$ of budget for the real amplifier (Part 2) and the (negligible, steady-state) startup leakage, comfortably under the $10\mu\text{A}$ total-current spec.

Component	Design value	Branch current	Role
R_1	600 k Ω	$I_1 = 1.150 \mu\text{A}$	CTAT leg (with Q1, unit BJT)
R_2	90 k Ω	$I_2 = 0.85 \mu\text{A}$	PTAT leg (with Q2, area = 24 $\times$)
R_3	400 k Ω	$I_1 + I_2 = 2 \mu\text{A}$	Output leg, sets $V_{OUT} = 0.8 \text{ V}$
N (Q2/Q1 area ratio)	24	–	Sets ΔV_{EB}

Table 3: Final Part-1 hand-calculated component values

2.4 PMOS Mirror Sizing (MP1, MP2, MP3)

I biased them in moderate inversion (target $g_m/I_D \approx 13 \text{ V}^{-1}$) using the ADT from the earlier labs, trading off r_o (want it high, for good current-source behavior and low sensitivity to V_{DS} mismatch we will set $L = 2\mu$)

The screenshot shows the Sizing Assistant interface. At the top, there is a 'State1' dropdown and a 'Save State' button. Below this, several input fields are visible: 'ID' set to '2u', 'gm/ID' set to '13', 'L' set to '2u', 'VDS' set to '1', 'VSB' set to '0', and 'Stack' set to '1'. There are 'Get' and 'Apply' buttons. Below these, the 'Y-Expr' is set to 'gm/ID*fT', with 'Plot' and 'Append' buttons. At the bottom, a 'Device Parameters' table is displayed.

#	Parameter	TT-27.0
1	ID	2u
2	IG	0
3	L	2u
4	W	10.4u
5	VDB	1
6	VGB	715.4m
7	VGS	715.4m
8	VDS	1
9	VSB	0
10	gm/ID	12.81
11	Vstar	156.1m

Figure 2: Sizing Assistant result for the core PMOS current mirrors

Parameter	Value
L	2u
W	10u
g_m/I_D	$\approx 13 \text{ V}^{-1}$

Table 4: MP1/MP2/MP3 sizing result

2.5 Startup Circuit Analysis (MP13, MN6, MN7)

How the startup circuit works:

- **MP13** (1/40, long channel) is a very weak, always-on pull-up. While the core is in the zero-current state, it injects a small leakage-level current into the core's shared bias node, nudging the loop away from the degenerate equilibrium.
- **MN7** (sense device) monitors that VREF has risen to a level once the loop starts up correctly.
- **MN6** then turns on and diverts/shorts the startup injection path, disengaging MP13 from the core so it contributes negligible current once the circuit is running – this is essential, since a startup circuit that keeps sinking current in steady state would eat directly into the $10 \mu\text{A}$ budget.

Sizing: I choose max L for the PMOS of the inverter and a width of 0.5u and for the NMOS of the inverter i picked a large width = 10u and L = 0.5u and for the sensing nmos i choose W = 5u and L = 1u as my first-pass starting point, since startup sizing is not directly coupled to the output-voltage/current spec – it just needs to be (a) strong enough to guarantee startup in the worst corner (SS, -40°C , where all currents and g_m are weakest), and (b) weak enough in steady state to not violate the current budget.

2.6 Behavioral Amplifier (VCVS) for Part 1

using VCVS with high gain $A \approx 60\text{--}80 \text{ dB}$.

2.7 Schematic with Device Sizing

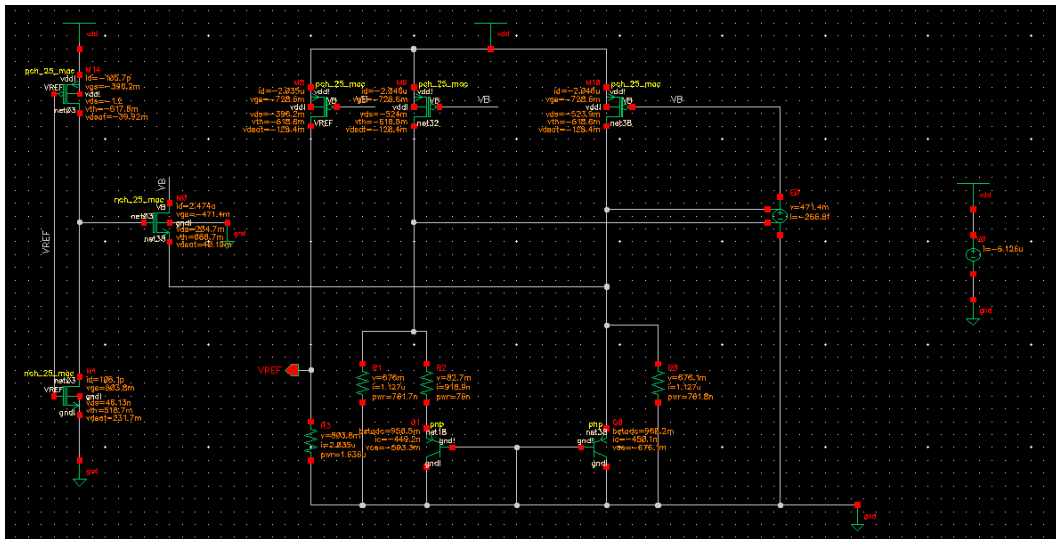


Figure 3: Part 1 schematic with device sizing

2.8 DC Operating Point

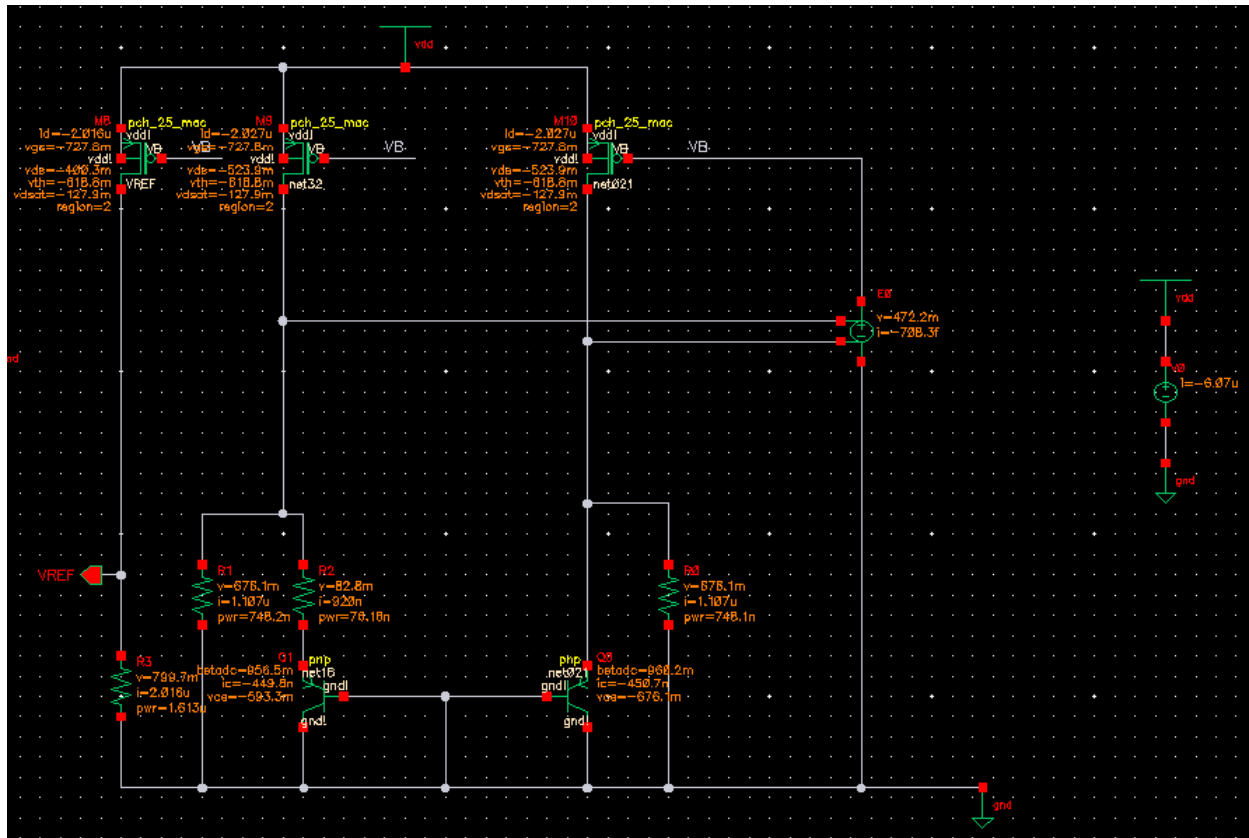


Figure 4: Part 1 – annotated DC operating point

Test	Output	Nominal	Spec	Weight	Pass/Fail
LUT_GEN:bgr_01:1	VDC("/VREF")	803.8m			

Figure 5: VREF Value

2.9 V_{OUT} vs. Temperature (All Corners)

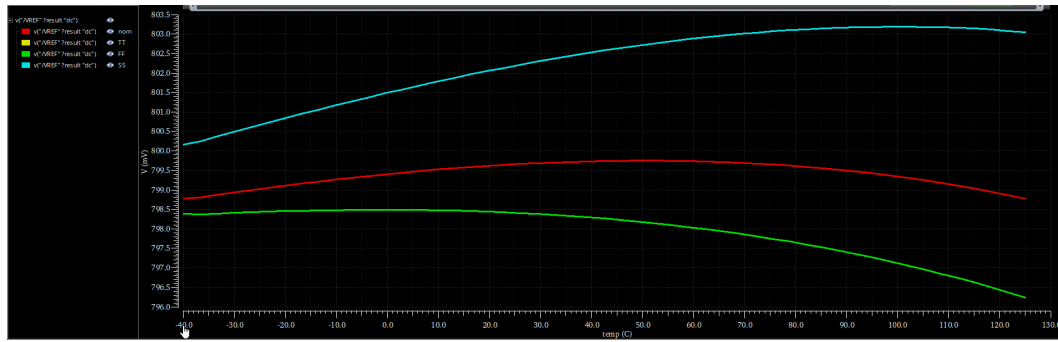


Figure 6: Part 1 – V_{OUT} vs. temperature across all process corners

Parameter		Nominal						TT	FF	SS
Save_All_OP.scs										
cm65gplus_2d5_1k_v...		tt_25						tt_bip	ff_bip	ss_bip
cm65gplus_2d5_1k_v...		tt_bip						tt_25	ff_25	ss_25
Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	TT	FF	SS
LUT_GEN:bgr_01:1	VDC("/VREF")	799.7m				798.4m	802.3m	799.7m	798.4m	802.3m
LUT_GEN:bgr_01:1	v("/VREF")?result "dc"									

Figure 7: Corner Simulation

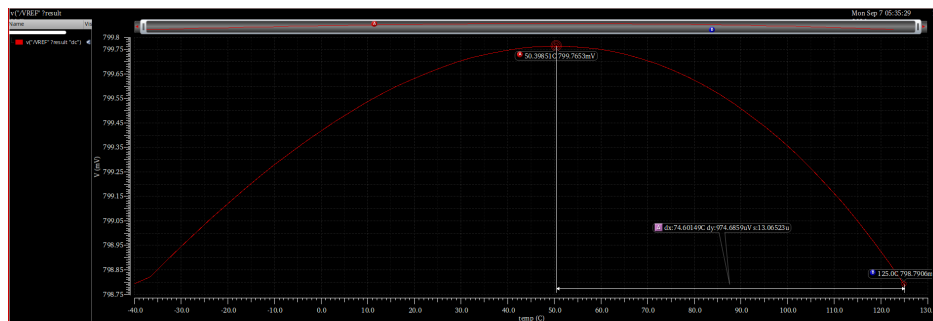


Figure 8: The load regulation

$$TC = \frac{V_{OUT,max} - V_{OUT,min}}{V_{OUT}(27^{\circ}C) \times \Delta T} \times 10^6 \text{ ppm}/^{\circ}C = 16.1 \text{ ppm}/^{\circ}C \quad (12)$$

3 Part 2: Self-Biased Amplifier (Replacing the Behavioral VCVS)

3.1 Amplifier Topology

i assumed that the current in every branch will equal 0.5u

- MN1, MN5 I implemented this pair using LVT device which is important here because the input common-mode level sits close to the low so i choose $W = 1\mu$ and $L = 5\mu$ for high gain.
- MN3, MN4 form the tail-current mirror biasing the input pair, sized using $L = 0.5\mu$ and g_m over $i_d = 12\mu$.
- MP5, MP6, MP7 form the PMOS current-mirror load sized using $L = 2\mu$ and g_m over $i_d = 13\mu$.

i needed to do some tuning to the circuit so that all mosfets in saturation.

3.2 Full Schematic with Device Sizing

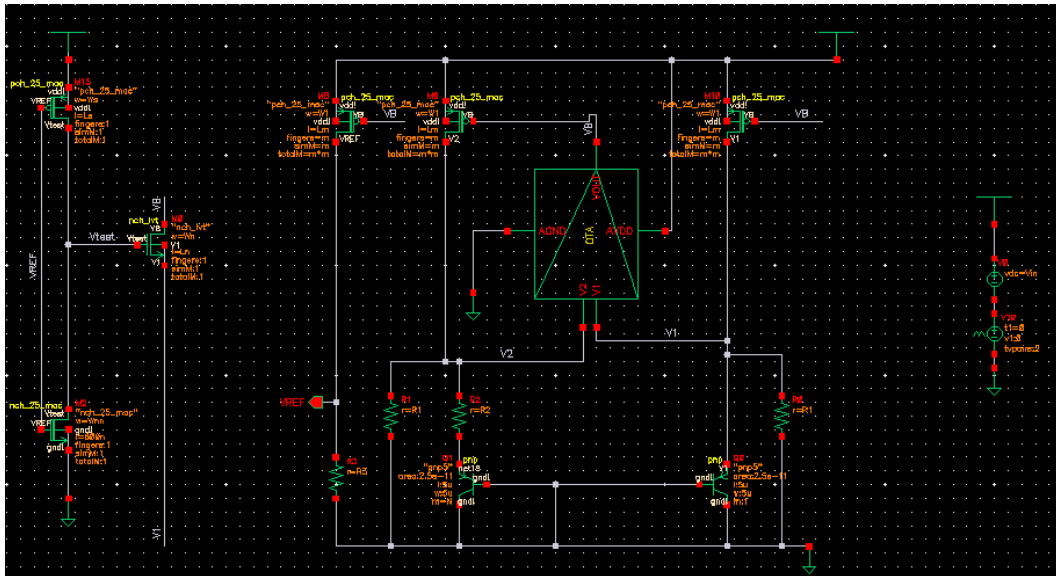


Figure 9: Part 2 schematic with device sizing

3.3 DC Operating Point

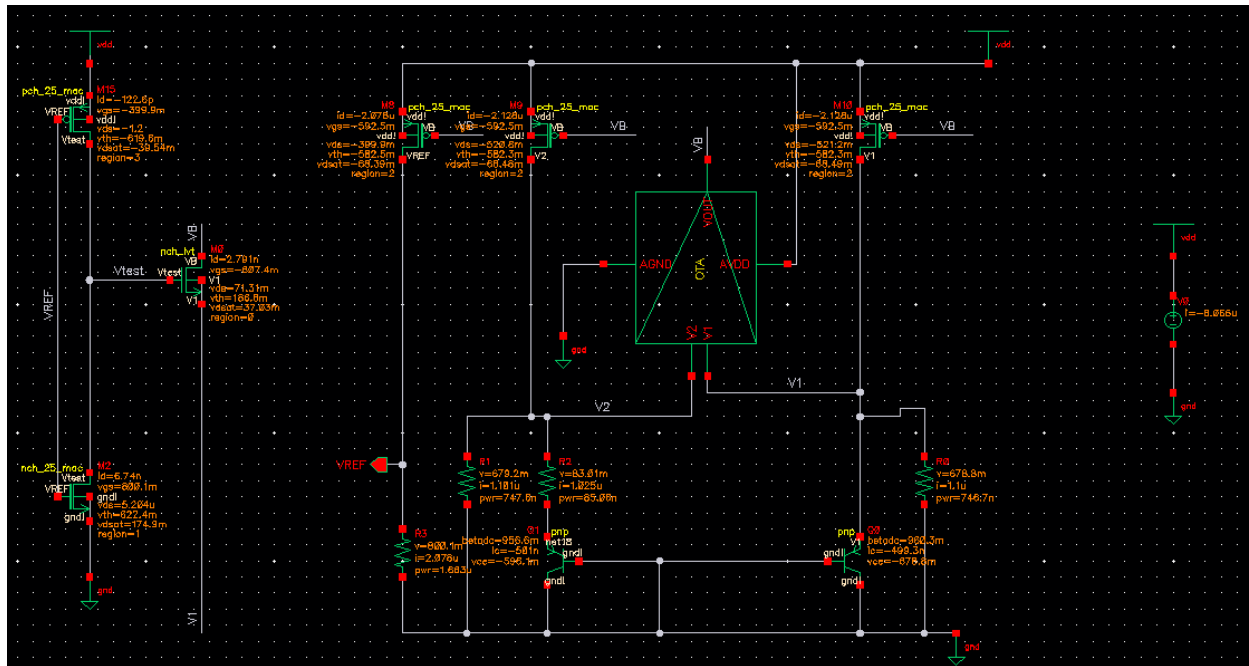


Figure 10: BGR annotated DC operating point

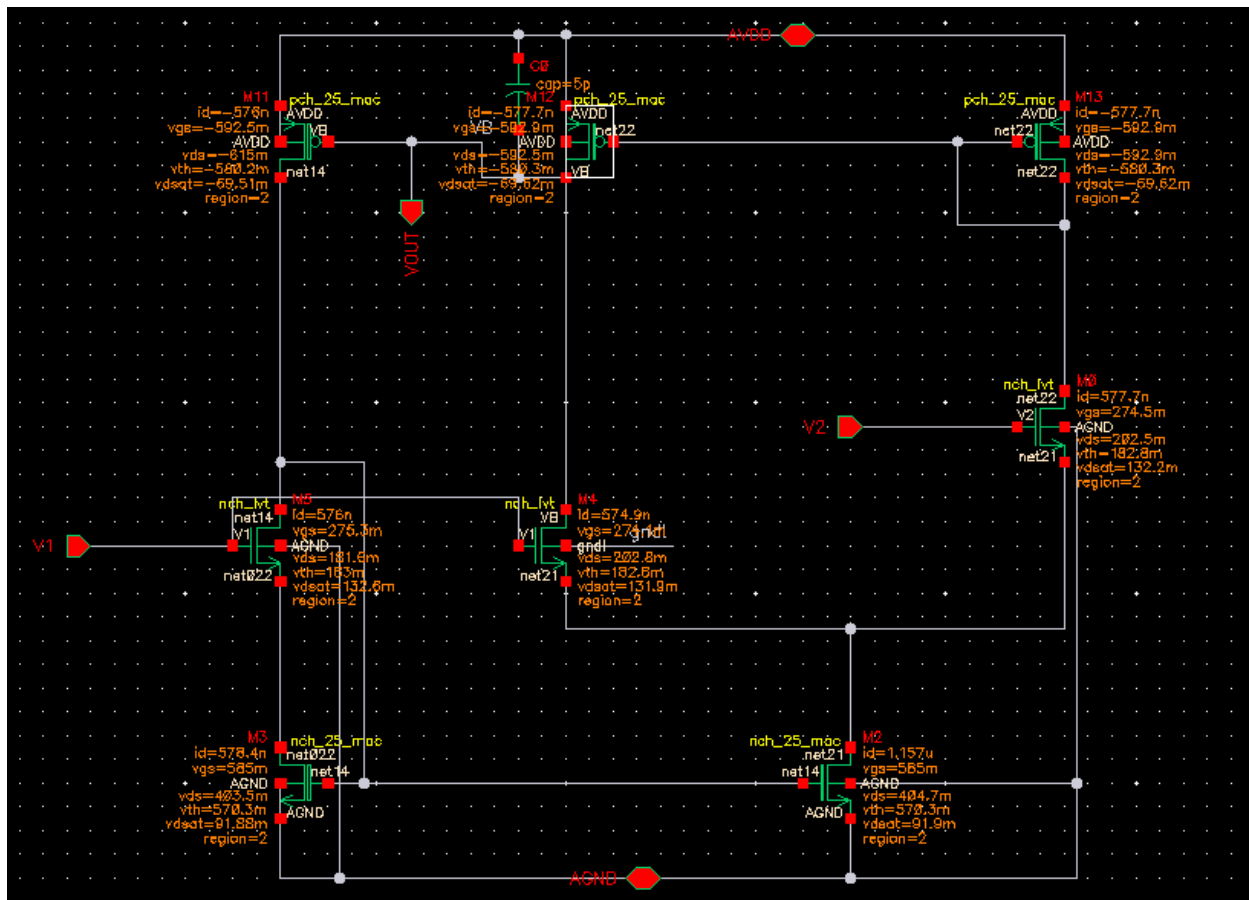


Figure 11: OTA annotated DC operating point

3.4 V_{OUT} vs. Temperature (All Corners)

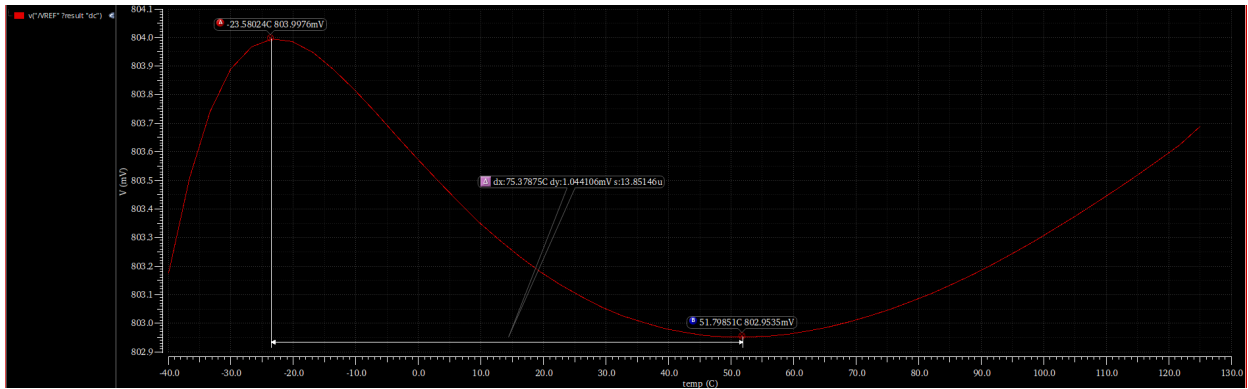
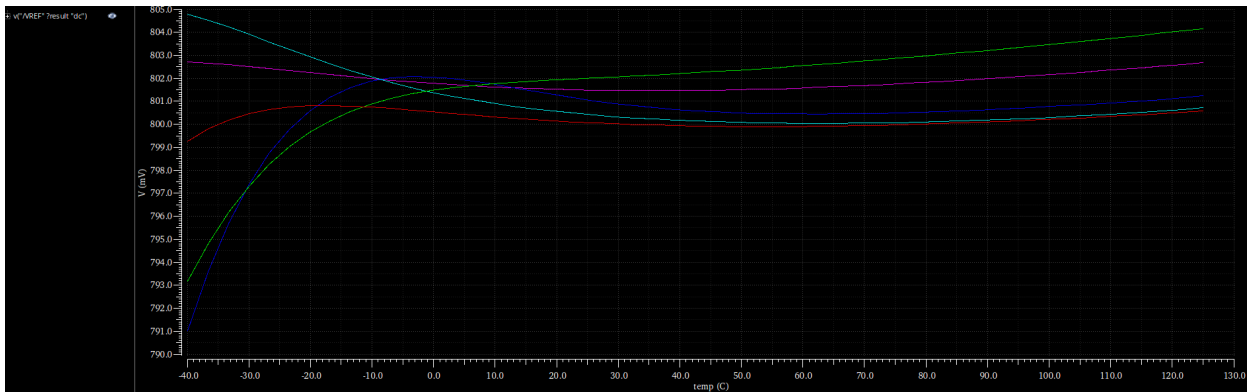


Figure 12: Load regulation

Figure 13: Part 2 – V_{OUT} vs. temperature across all process corners

3.5 Loop-Stability (STB) Analysis

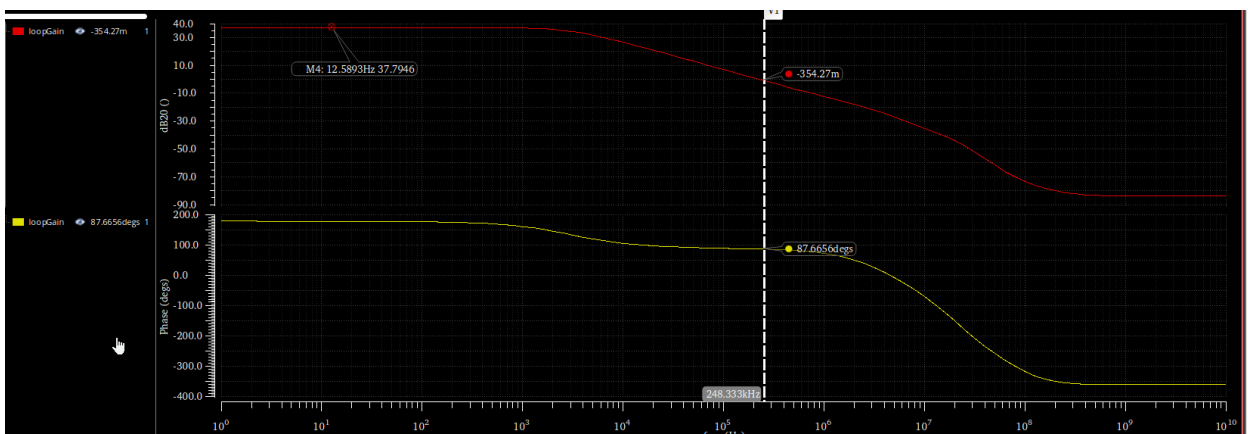


Figure 14: Loop-gain STB analysis: magnitude and phase vs. frequency

$$PM = 88$$

3.6 Transient Simulation – V_{DD} Ramp

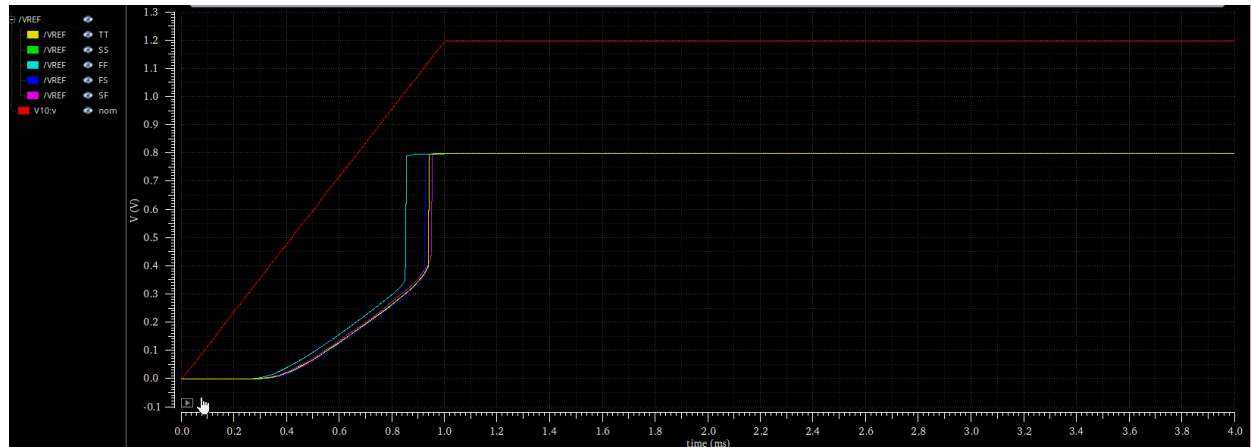


Figure 15: Transient response of V_{OUT} during power-up ramp

4 Part 3 (Bonus): Real Passives from the PDK

4.1 Passive Selection

4.2 DC Operating Point

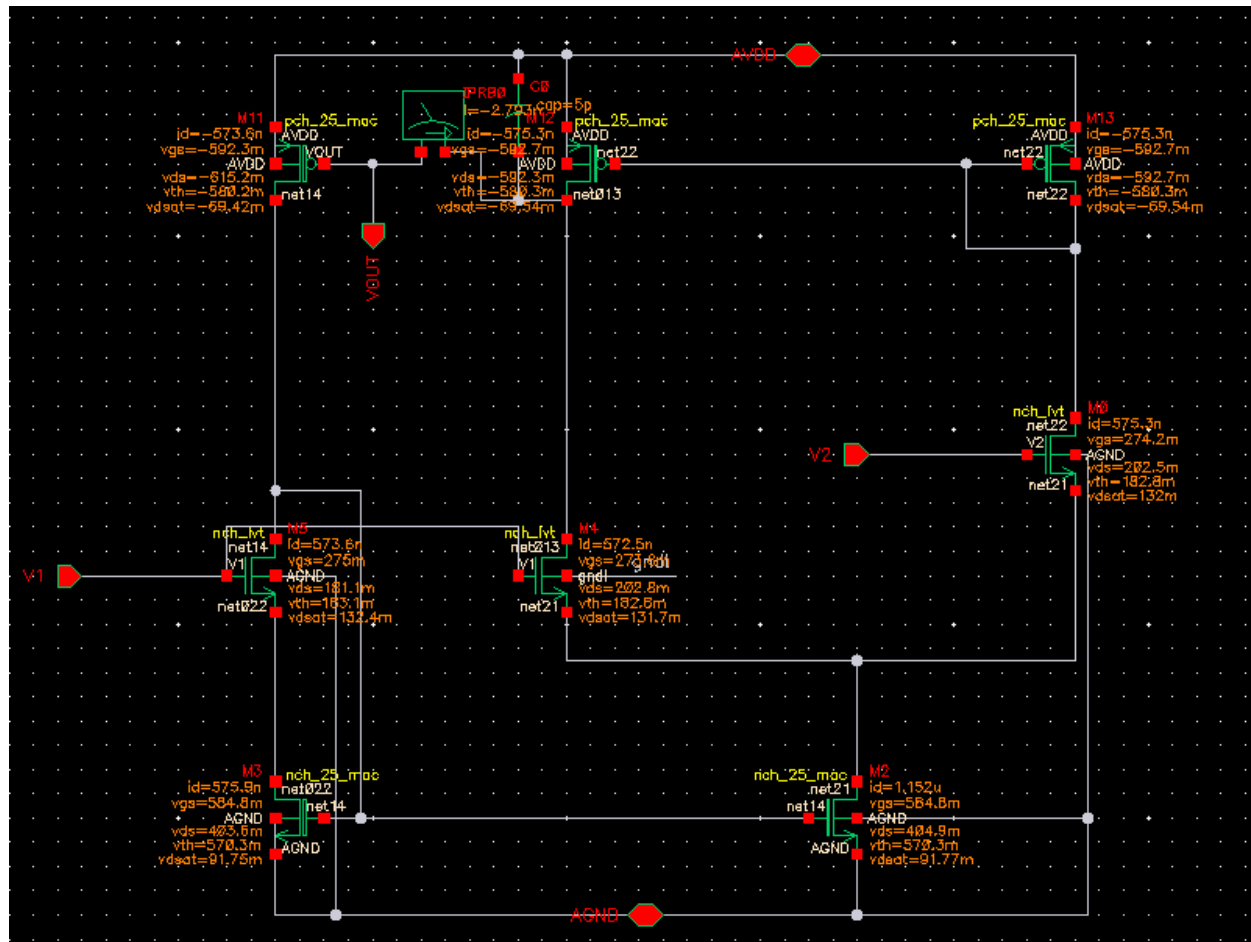


Figure 16: BGR annotated DC operating point

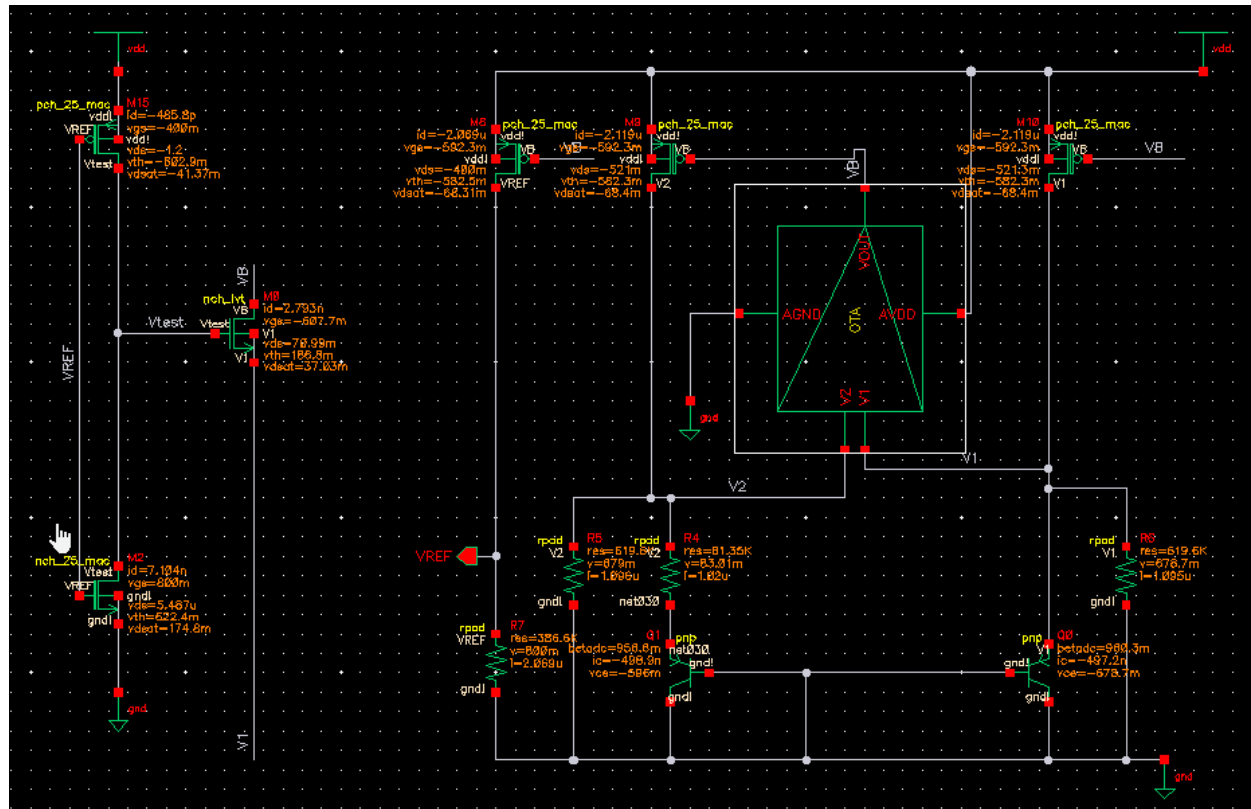
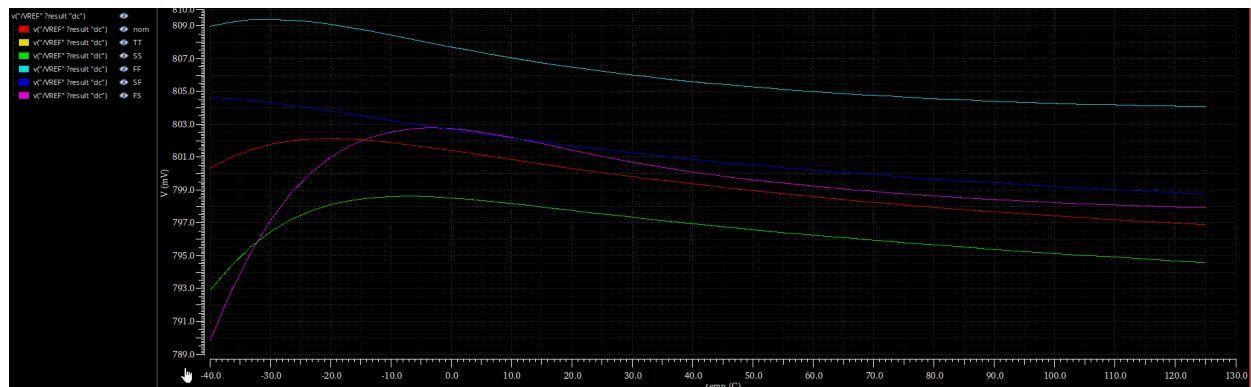


Figure 17: OTA annotated DC operating point

4.3 V_{OUT} vs. Temperature (All Corners)

Figure 18: Part 3 – V_{OUT} vs. temperature with real PDK passives

5 Summary

Metric	Spec	Part 1 (ideal amp)	Part 2 (real amp)
V_{OUT} @ 27°C, TT	0.800 V	0.7997	0.801
Total bias current	< 10 μ A	6.1u	8.1u
Phase margin (worst corner)	> 60°	N/A (ideal amp)	88

Table 5: Final results summary against the design specs